



A
500 I

u

B
200 II

8 A u 2 B

2 A u 8 B

U:

2 A

3 B

A
x

I

B
500-x

8 x

2(500-x)

y

200-y

2 y

8(200-y)

A: $8x + 2y$

B: $2(500-x) + 8(200-y)$

$$3(8x + 2y) = 2(500-x) + 8(200-y)$$

$$24x + 6y = 1000 - 2x + 1600 - 8y$$

$$24x + 2x = 2600 - 14y$$

$$x = \frac{2600 - 14y}{26}$$

$$f(x; y) = 8x + 2y =$$

$$8 \left(\frac{2600 - 14y}{26} \right) + 2y = f(y)$$

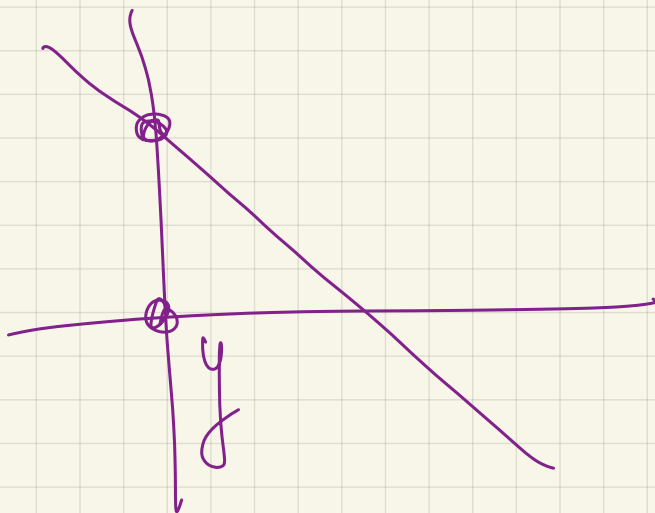
$$800 - \frac{8 \cdot 14y}{26} + 2y$$

$$800 - \frac{4 \cdot 14y}{13} + 2y = f(y)$$

$$800 - \frac{4 \cdot 14y + 2 \cdot 13y}{13} = f(y)$$

$$f'(y) < 0 \Rightarrow f(y) \text{ убывает}$$

$$\begin{aligned} y &= 0 \\ x &= 100 \end{aligned}$$



800 штук!

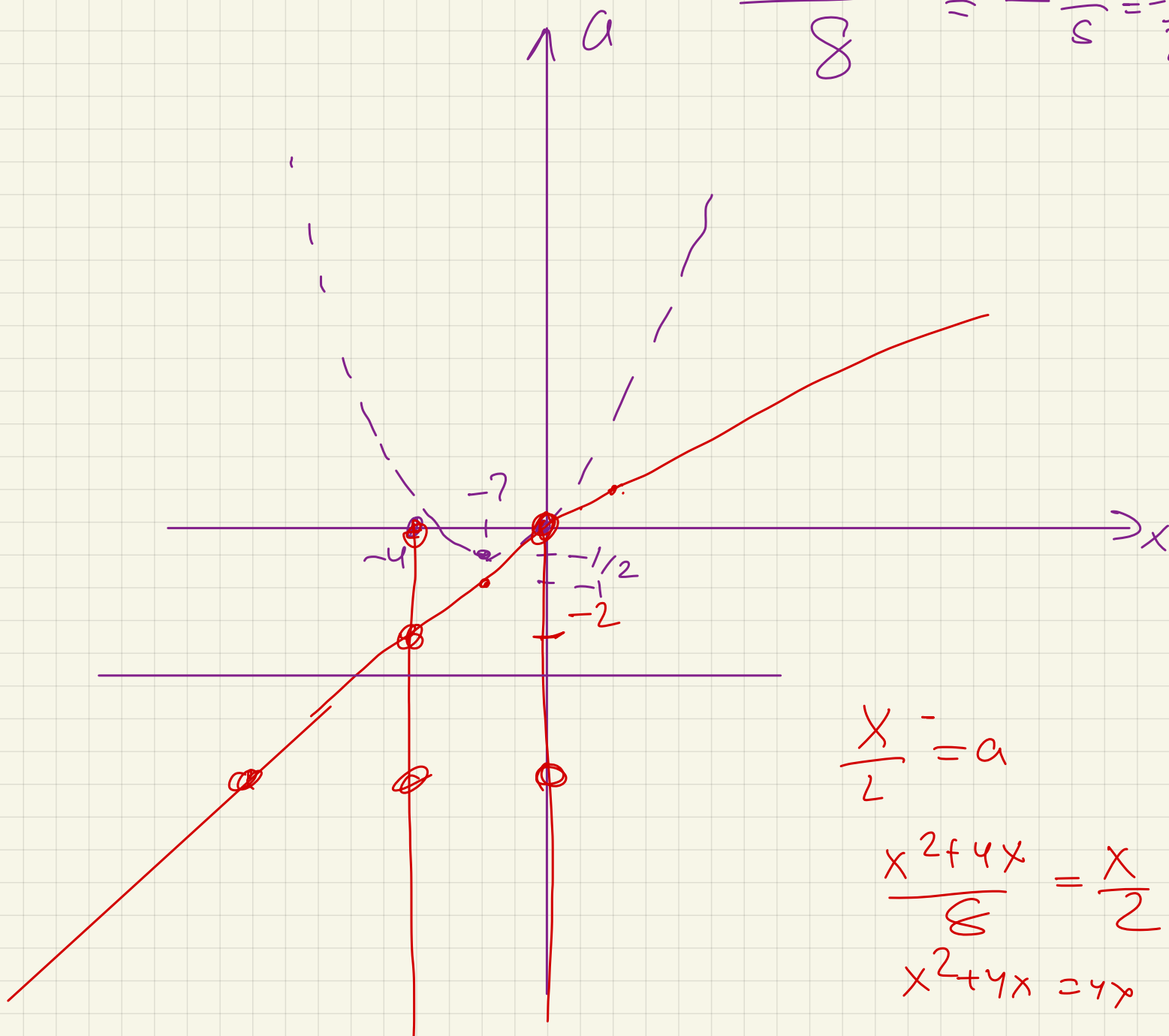
$$x=0$$

$$\begin{cases} x(x-2a)(x+4)=0 \\ x^2+4x-8a \geq 0 \end{cases}$$

3 cases

$$a \leq \frac{x^2+4x}{8}$$

$$\frac{4-8}{8} = -\frac{4}{8} = -\frac{1}{2}$$



$$\frac{x}{2} = a$$

$$\frac{x^2+4x}{8} = \frac{x}{2}$$

$$x^2+4x=4x$$

$$a \in (-\infty; -2) \cup (-2; 0)$$

$$x^2 = 0$$

$$x = 0$$

$$f(x) = 3ax^4 - 8x^3 + 3x^2 - 7$$

$$a \geq 0$$

$$[-1; 1]$$

$$1 \text{ T. mem}$$

$$f'(x) = 12ax^3 - 24x^2 + 6x = 0$$

$$6x(2x^2a - 4x + 1) = 0$$

$$x = 0$$

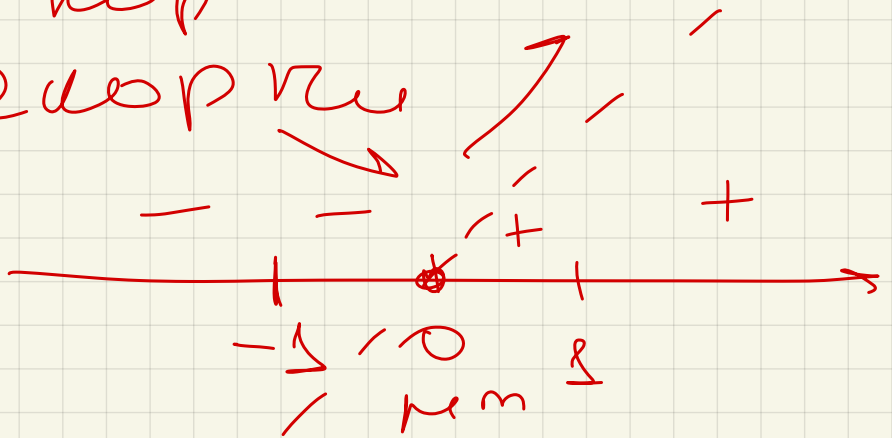
$$a \geq 0$$

$$1) a < 0 \rightarrow 0 \text{ roots}$$

$$2) a = 0 \rightarrow 1 \text{ root}$$

$$3) a > 0 \rightarrow 2 \text{ or } 3 \text{ roots}$$

1)



$$4^2 - 4 \cdot 2a < 0$$

$$16 - 8a < 0$$

$$a > \frac{16}{8} = 2$$

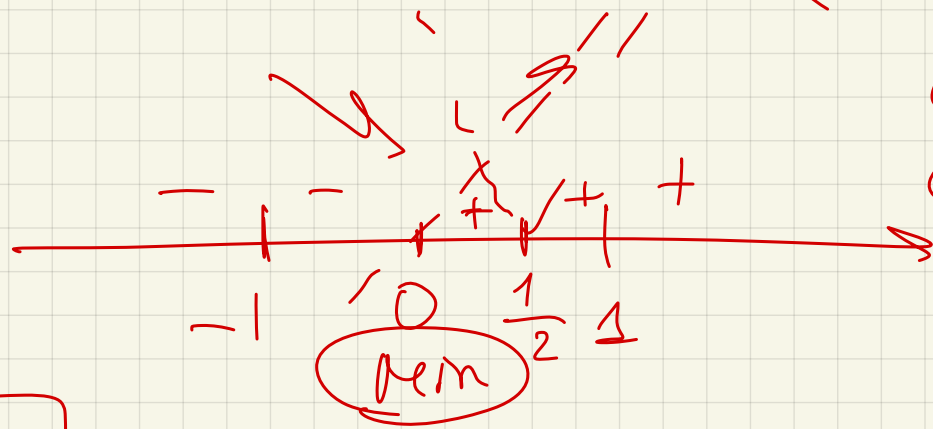
$$a > 2$$

$$2) \quad D = 0$$

$$a = 2$$

$$4x^2 - 4x + 1$$

$$(2x - 1)^2$$



$$x = \frac{1}{2}$$

$$x = 0$$

$$a = 2$$

$$3) \quad D > 0 \Rightarrow 2 \text{ roots}$$

$$2ax^2 - 4x + 1 = 0$$

$$\frac{4 \pm \sqrt{4^2 - 4 \cdot 2a}}{4a} = \frac{4 \pm 2\sqrt{4 - 2a}}{4a}$$

$$= \frac{2 \pm \sqrt{4 - 2a}}{2a}$$

$$x_1 = \frac{2 - \sqrt{4 - 2a}}{2a}$$

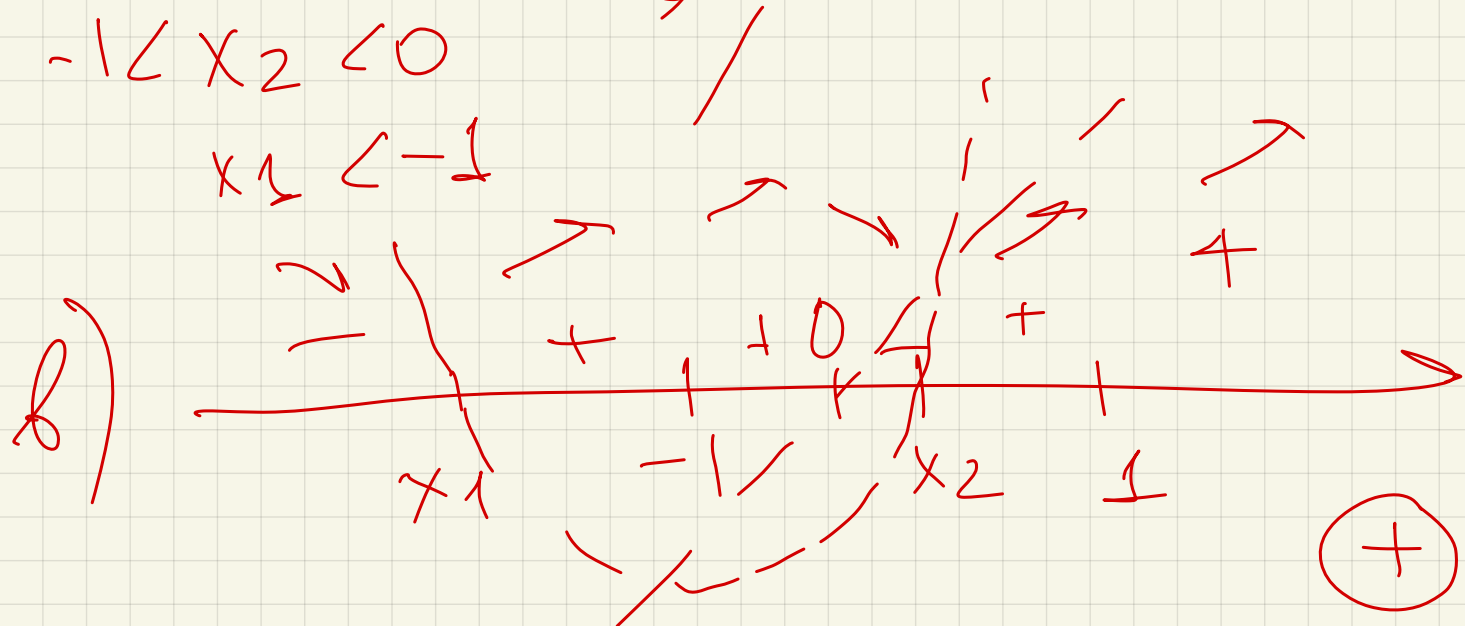
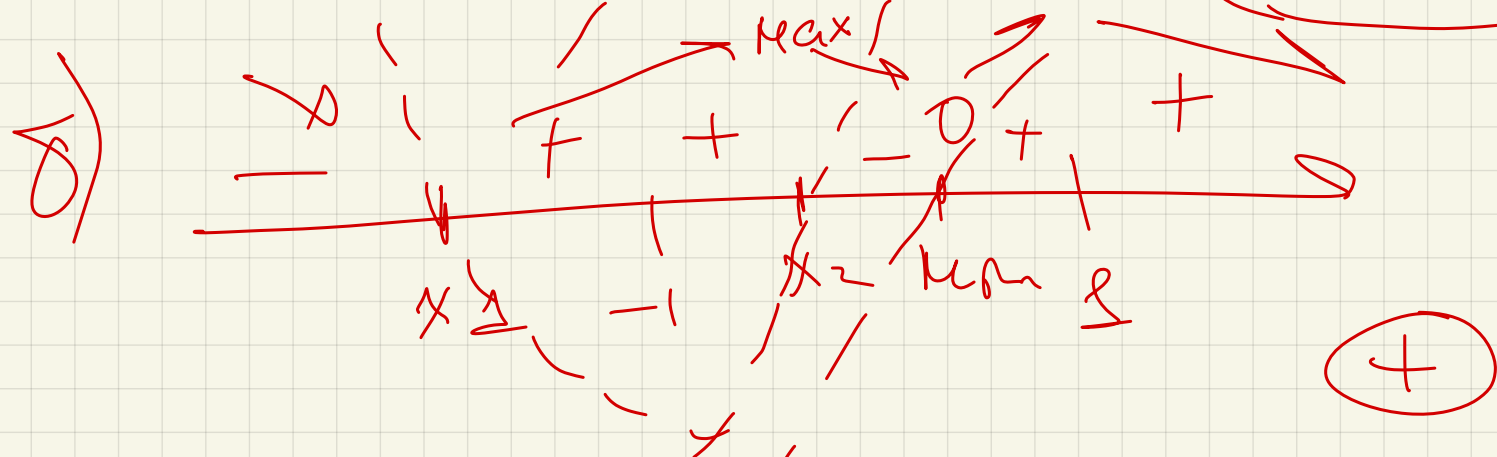
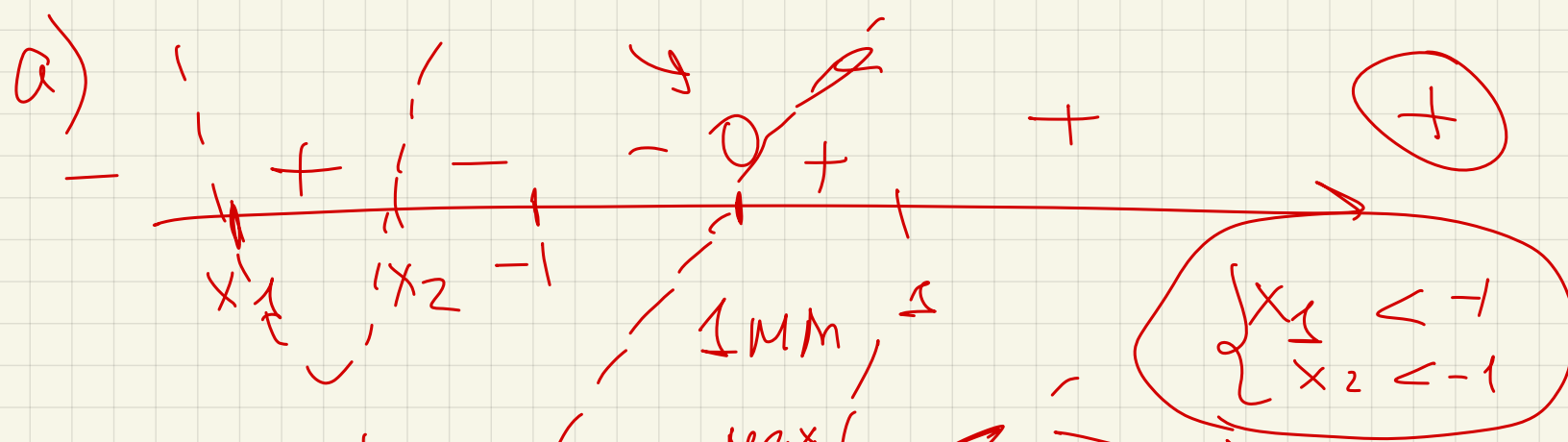
$$x_2 = \frac{2 + \sqrt{4 - 2a}}{2a}$$

$$4 - 2a > 0$$

$$a < 2$$

$$a \geq 0$$

$$(2ax^2 - 4x + 1)$$

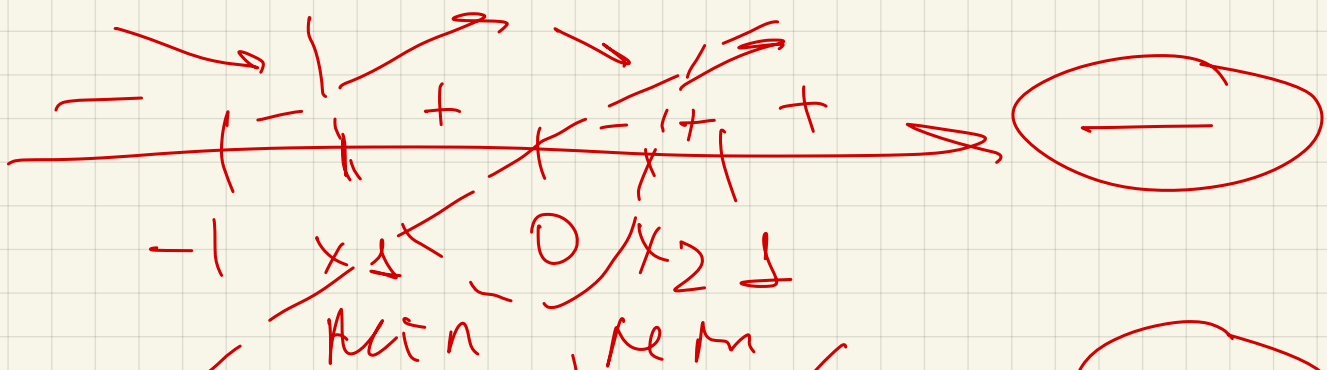


But $-1 < x_2 < 1$
 $x_1 < -1$

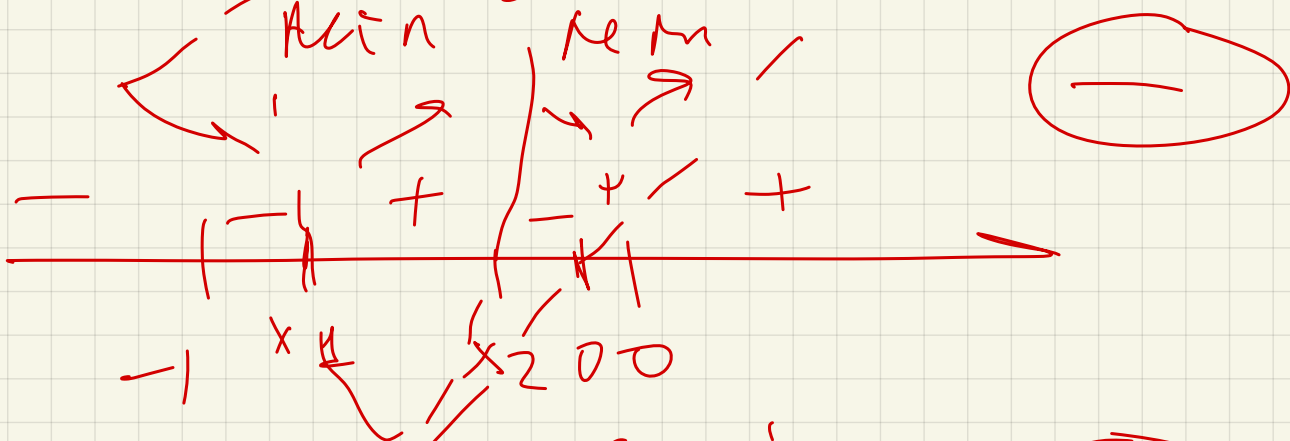
$\begin{cases} x_1 < -1 \\ x_2 < 1 \end{cases}$

$\begin{cases} x_1 < -1 \\ -1 < x_2 < 1 \end{cases}$

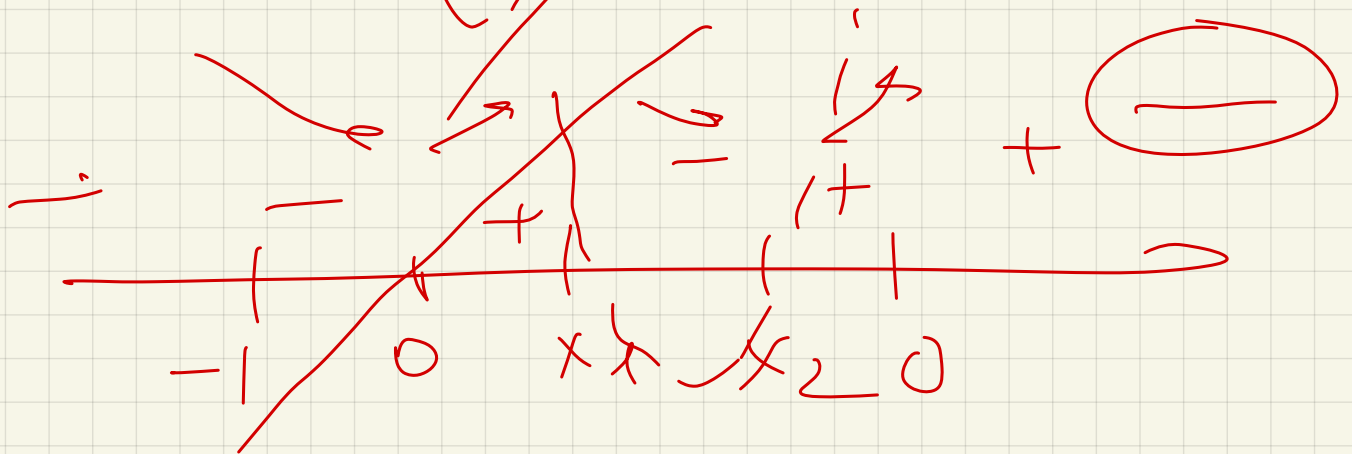
2)



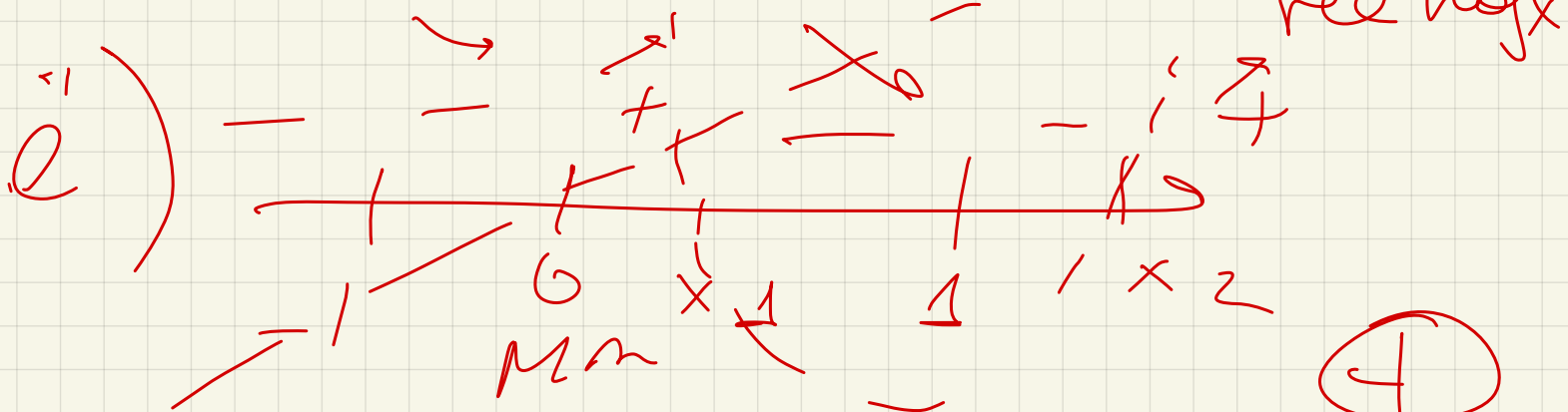
g)



z)

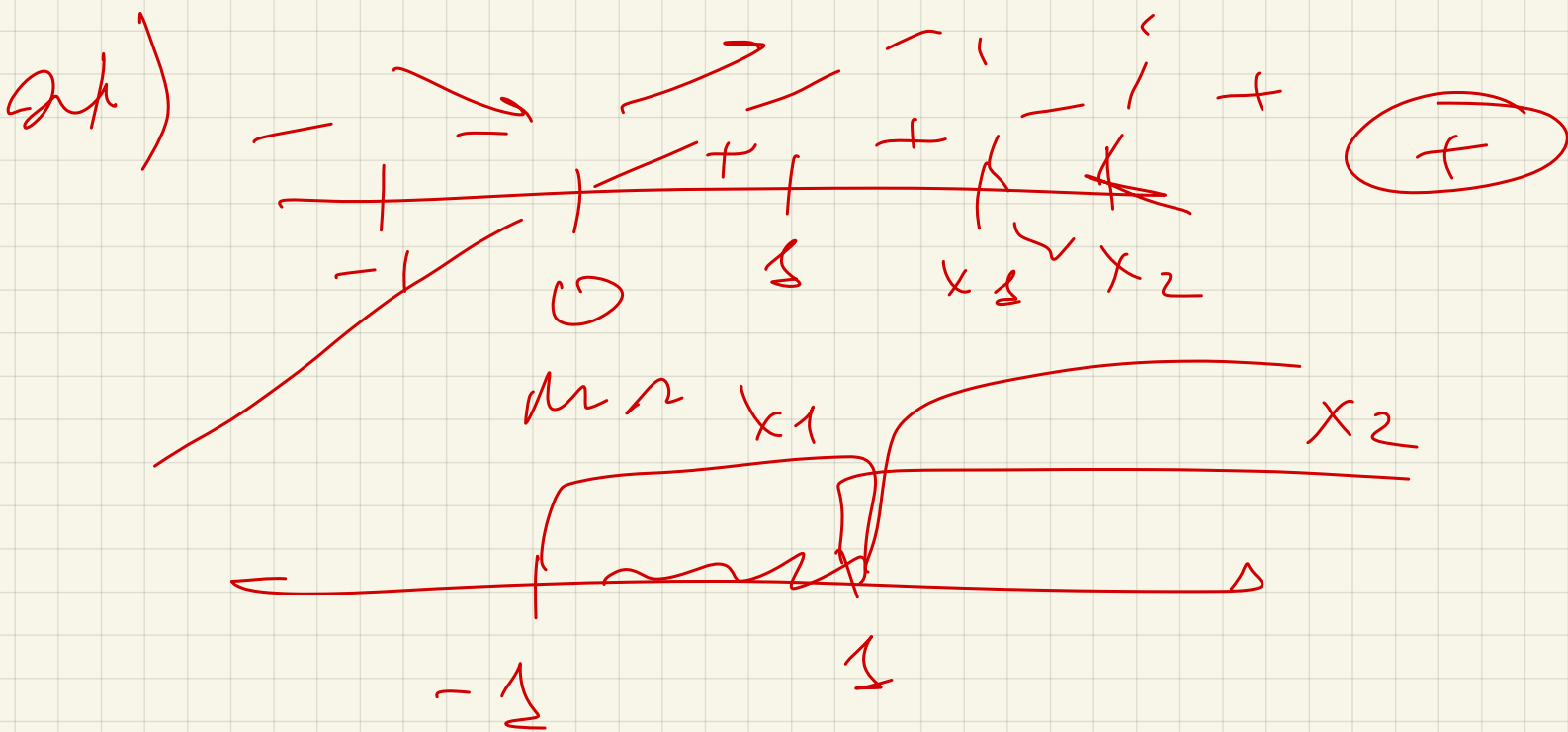


одна корна на $[-1; 1]$



$$-1 < x_1 < 1$$

$$x_2 > 1$$



$\textcircled{V} \left\{ \begin{array}{l} x_1 > -1 \\ x_2 > 1 \end{array} \right. \quad \text{a mesor}$
 $\left\{ \begin{array}{l} x_1 < -1 \\ x_2 < 1 \end{array} \right. \quad \emptyset \quad \emptyset$

$$x_1 = \frac{2 - \sqrt{4 - 2a}}{2a}$$

$$x_2 = \frac{2 + \sqrt{4 - 2a}}{2a}$$

$$\frac{(2 - \sqrt{4 - 2a})}{2a} > -1$$

$$\frac{2 + \sqrt{4 - 2a}}{2a} > 1$$

$$2 + \sqrt{4 - 2a} > 2a$$

$$\rightarrow a < 0$$

$$a < \frac{3}{2}$$

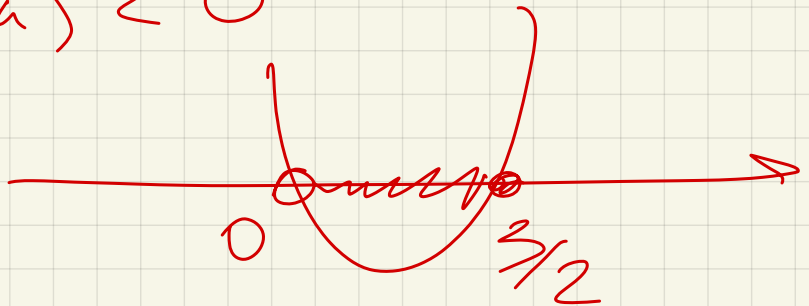
$$4 - 2a > (2a - 2)^2$$

$$4a^2 - 8a + 4 - 4a + 4 < 0$$

$$4a^2 - 6a < 0$$

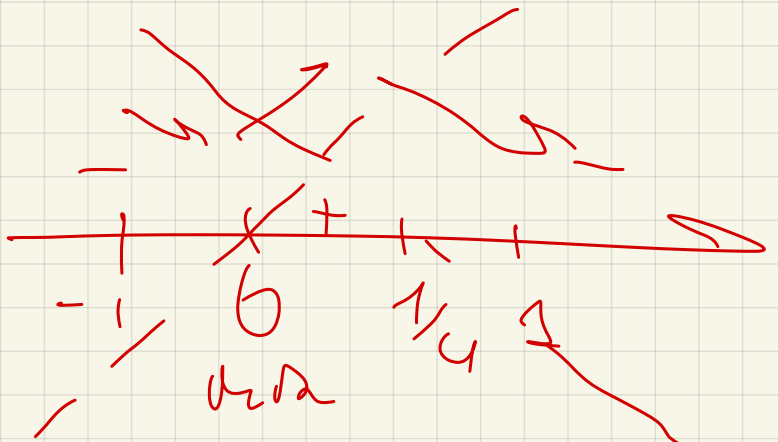
$$2a(2a - 3a) < 0$$

$$a \in \left(0; \frac{3}{2}\right)$$



$$a = 0$$

$$6x(-4x + 1)$$



$$a \in [0; \frac{3}{2}) \cup [2; +\infty)$$

$$y = a|x| + |a| - 3$$

$$y = \frac{a}{3}$$

$$a > 0$$

$$A < 11$$

$$a - 3 \geq \frac{a}{3}$$

$$a - \frac{a}{3} < 3$$

$$\frac{2a}{3} < 3$$

$$0 < a < \frac{9}{2}$$

$$\frac{a}{3} - (a - 3) =$$

$$\frac{a - 3a + 9}{3} =$$

$$\frac{9 - 2a}{3} = A11$$

$$3a + 3a - 9 = a$$

$$x = \frac{-2a + 9}{3a}$$

$$X = \frac{2a-9}{3a}$$

$$\frac{-2a+9}{3a} - \left(\frac{2a-9}{3a} \right) =$$

$$\frac{-2a+9-2a+9}{3a} = \frac{18-4a}{3a}$$

$$BC = \frac{18-4a}{3a}$$

$$\frac{AM \cdot BC}{2} \geq \frac{1}{3}$$

$$\frac{1}{2} \left(\frac{9-2a}{3} \right) \left(\frac{18-4a}{3a} \right) \geq \frac{1}{3}$$

$$(9-2a)(18-4a) \geq \frac{2(9-2a)^2}{2 \cdot 3a} \geq 1$$

$$(9-2a)^2 \geq 3a$$

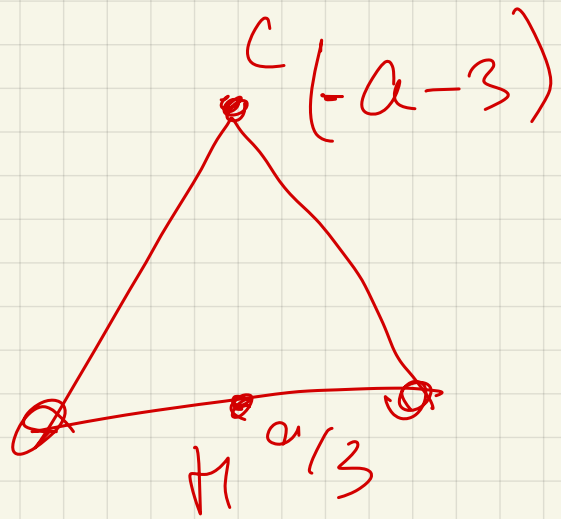
$$81 - 36a + 4a^2 - 3a \geq 0$$

$$|a^2 - 39a + 8| \geq 0$$

$$a > 0$$

$$a < 4,5$$

$$x = \frac{2a+9}{-3a}$$



$$C > H$$

$$-a-3 > \frac{a}{3}$$

$$-3 > \frac{a}{3} + a$$

$$a|x| - a - 3 \Rightarrow$$

$$\frac{4a}{3} < -3$$

$$a < -\frac{9}{4}$$

$$ax - a - 3 = \frac{a}{3}$$

$$-ax - a - 3 = \frac{a}{3}$$

$$3ax = a + 9 + 3a$$

$$x = \frac{4a+9}{3a}$$

$$-3ax - 3a - 9 = a$$

$$-3ax = 4a + 9$$

$$x = \frac{4a+9}{-3a}$$

$$\frac{1a+9}{3a} + \frac{4a+9}{3a}$$

$$BC = \frac{8a+18}{3a}$$

$$AH = -a - 3 - \frac{a}{3} =$$

$$\frac{-4a-9}{3}$$

$$\frac{1}{2} \left(\frac{8a+18}{3a} \right) \left(\frac{-9-4a}{3} \right) \geq \frac{1}{3}$$

$$a < -\frac{9}{4}$$

$$-\frac{2(4a+a)^2}{2 \cdot 3a} \geq 1$$

$$-\frac{(4a+9)^2}{3a} \geq 1$$

$$\left\{ \begin{array}{l} \frac{(4a+9)^2}{3a} \leq -1 \\ a < -\frac{9}{4} \end{array} \right.$$